ETW3482 Week 6 Tutorial

Section 1 Demo: Creating a Week 6 pipeline

Purpose: Become familiar with the Logistic Regression node and observe how model-selection methods, effect options, and the binary classification cutoff influence the fitted model and its results.

We will create the demonstration pipeline from the CPML Demo template saved in Exchange

Note: This is the task shown in the demonstration video. However, keep in mind that SAS Viya uses distributed processing, so the values in the results will vary slightly across runs.

1. Open the project named Demo in SAS Model Studio.
2. Create a new pipeline and name it Week 6.
3. Select the saved CPML demo template as the starting template and create the pipeline.
4. Check that the Logistic Regression node appears immediately after the Variable Selection node and is connected correctly.
5. Rename the existing Logistic Regression node LR - Stepwise Baseline.

Section 2 Demo: Inspect the general properties and build the baseline model

Use the existing node to identify the important property groups before changing any settings.

1. Select LR - Stepwise Baseline and inspect its Properties pane. Expand the following groups:

    a. General properties

Property	Setting for this demonstration
Binary target link function	Keep Logit. It links the log odds of churn to a linear combination of the selected effects.
Class input order	Keep the default unless a specific reference ordering is needed.
Class input coding	Keep GLM coding. The estimated class-level effects are interpreted relative to a reference level.
Interpretation reminder: A positive coefficient increases the log odds, and therefore usually the predicted probability, of the event level when the corresponding input increases or when the stated class level is present. The direction must always be interpreted relative to the target event and the class-variable reference level.	

    b. Effects options, Selection method and Selection options

Property path: Effects Options and Selection Method / Selection Options	
Property	**Setting for this demonstration**
Two-factor interactions	Deselected
Polynomial	Deselected
Selection method	Stepwise
Effect-selection criterion	Significance level
Selection-process stopping criterion	Significance level
Model-selection criterion	Validation ASE
Entry significance level	0.05
Stay significance level	0.05

2. Run LR - Stepwise Baseline. When the run is complete, right-click the node and open Results.
3. Review the following outputs:
    a. **Selection Summary:** shows which effects entered or left at each selection step.
    b. **Parameter Estimates and t Values by Parameter:** show coefficient direction, uncertainty, and statistical evidence for the final effects.
    c. **Regression Fit Statistics and Fit Statistics:** compare training, validation, and test performance where available.
    d. **ROC Reports:** show discrimination over the full range of probability cutoffs.
    e. **Event Classification:** shows the confusion matrix at the available cutoffs.

> **Do not conclude yet:** The baseline result provides a reference. A lower training error alone does not establish that another option is better; prioritise validation performance and the model's usefulness for the churn decision.

Section 3 Demo: Compare Stepwise, Forward, and Backward selection

Change only the selection direction so the comparison remains interpretable.

1. Duplicate LR - Stepwise Baseline twice. Rename the duplicate nodes LR - Forward and LR - Backward.

Selection method	How the search proceeds
LR - Stepwise Baseline	Starts with no candidate effects, adds effects, and can later remove effects that no longer satisfy the stay condition.
LR - Forward	Starts with no candidate effects and adds effects. Effects already entered are not removed by the forward procedure.
LR - Backward	Starts with all candidate effects and removes effects until the stopping condition is met.

2. For each duplicate node:
    a. Change Selection method to the method shown in the node name.
    b. Keep Effect-selection criterion as Significance level.
    c. Keep Selection-process stopping criterion as Significance level.
    d. Keep Model-selection criterion as Validation ASE.
    e. Keep Entry significance level at 0.05 where it applies and Stay significance level at 0.05 where it applies.
    f. Keep all effect options and the classification cutoff identical to the baseline model.
3. Run the three selection nodes. For each node, examine the Selection Summary, the final Parameter Estimates, and Validation ASE. Consider:
    a. Do the three methods retain the same effects?
    b. Which method produces the simplest final model?
    c. Does the model with more effects achieve better validation performance?
    d. Are training and validation results diverging, suggesting possible overfitting?
4. Refer to the Slido and answer the question.

    How the criteria work together: The significance-level settings guide the entry and removal of effects during the search. Validation ASE is then used to choose the optimal model from the models visited during that search. Smaller Validation ASE indicates better validation prediction according to this criterion.

    Fair comparison: Selection method is the experimental change in this section. Do not change the effect set, significance levels, partitions, or cutoff while comparing the three search directions.

Section 4 Demo: Compare main effects, interactions, and quadratic effects

Create separate nodes so that each expansion of the model space can be assessed against the same baseline.

Two-factor interactions

1. Duplicate LR - Stepwise Baseline and rename the duplicate LR - Interactions.
2. Select Two-factor interactions under Effects Options. Keep all other options in Effect Options
3. Under Selection Options, set Hierarchy to All variables for this demonstration. This keeps the relevant main effects in the model whenever an interaction involving them is retained.

   Interpretation: An interaction means that the effect of one input depends on the level of another input. Do not interpret an interaction coefficient as though it were an independent main effect.

Quadratic effects

4. Duplicate LR - Stepwise Baseline again and rename the duplicate LR - Polynomial 2.
5. Select Polynomial under Effects Options. Keep all other options in Effect Options
6. Under Selection Options, set Hierarchy to All variables so a retained squared term is accompanied by its main effect.
7. Run LR - Interactions and LR - Polynomial 2. Compare both with LR - Stepwise Baseline.
8. Compare Validation ASE and other validation statistics using the same data partition.
9. Examine the final effect list to determine whether interaction or quadratic effects survived selection.
10. Check whether any improvement on training data is also present on validation data.
11. Prefer the simpler main-effects model when the expanded models do not provide a meaningful validation improvement.
12. Refer to the Slido and answer the question.

    Important: Polynomial and spline effects cannot be selected together. The spline option is not used in this demonstration.

    Why separate nodes are used: Each node changes one effect structure while preserving the remaining settings. Combining interactions and quadratic effects in one node would make it difficult to identify which expansion influenced model performance.

Section 5 Demo: Test a binary classification cutoff of 0.30

Separate probability estimation from the decision rule used to convert a probability into a predicted class.

1. Duplicate LR - Stepwise Baseline and rename the duplicate LR - Cutoff 0.30.
2. Select Specify node binary classification cutoff under Binary Classification Cutoff
3. Enter 0.30 for Node binary classification cutoff.
4. Do not change any other property.
5. Run LR - Cutoff 0.30 and open its Results.
6. Refer to the Slido and answer the question.
7. Compare the Event Classification results for LR - Stepwise Baseline and LR - Cutoff 0.30. When the churn event is coded correctly, lowering the cutoff from 0.50 to 0.30 generally produces the following changes

Result	Expected direction after lowering the cutoff
Predicted churn cases	Increase or remain the same
True positives and sensitivity/recall	Increase or remain the same
False negatives	Decrease or remain the same
False positives	Increase or remain the same
True negatives and specificity	Decrease or remain the same
Precision and overall accuracy	Can increase or decrease, depending on the data and class distribution
Key distinction: The cutoff does not change the fitted coefficients or posterior probabilities. It changes only how a probability is translated into a predicted class. Therefore, the ROC curve and probability-based measures such as AUC should not change merely because the cutoff changes.	
SAS output: The node-defined cutoff is included in the Event Classification report alongside the default 0.50 cutoff and the KS cutoff. The user-defined cutoff affects threshold-dependent statistics, including the misclassification rate at the user-defined cutoff.	

Discuss whether a cutoff of 0.30 is preferable for the churn problem. The answer depends on the consequences of missing a likely churner compared with the cost of contacting a customer who would not have churned.

8. Model comparison record

Model	Key change	Validation ASE	AUC or KS	Selected effects	Comment
LR - Stepwise Baseline	Main effects; Stepwise				
LR - Forward	Forward selection				
LR - Backward	Backward selection				
LR - Interactions	Nominal x nominal interactions				
LR - Polynomial 2	Quadratic interval effects				

Do not rank the cutoff node as a new predictive model: LR - Cutoff 0.30 has the same fitted probability model as its baseline duplicate. Evaluate it as an alternative decision rule, not as a model that has learned a different relationship from the data.

Brief introduction: GLM and Linear Regression nodes

The current demonstration uses Churn Flag as a binary class target. For this reason, the Logistic Regression node is the appropriate regression-family node for the practical demonstration.

Node	Target accepted in this SAS node	Core model	Use in this demo
Logistic Regression	Binary or nominal class target	Models event odds or class probabilities using an appropriate link function; Logit is used here.	Demonstrated
GLM	Interval target metadata	Extends the linear model by allowing a selected target distribution and a link between the response mean and the linear predictor.	Introduced only
Linear Regression	Interval target	Uses least squares and assumes a normal response with constant variance; predicts a continuous value.	Introduced only

Why GLM is not demonstrated: Although a generalised linear model can conceptually represent several response distributions, the GLM node documented for this SAS Viya version requires the target variable to have interval metadata. The current Churn Flag target is defined for classification, so it is not used in a GLM-node demonstration.

Why Linear Regression is not demonstrated: Linear Regression requires an interval target and produces unrestricted continuous predictions. Applying it directly to a binary class target would not provide the classification workflow and probability interpretation intended for this churn problem.